

Project Description

- Data Understanding:

This project focuses on predicting customer churn in a telecommunications company using the Telco Customer Churn dataset. Each record represents an individual customer, and the dataset contains 7043 observations with 21 features describing customer demographics, services, account information, and billing details.

- Objective:

The primary goal is to build a classification model that predicts whether a customer is likely to churn (leave the company) based on historical data. This enables the business to take proactive retention actions.

- Target Variable:

****Churn****: Indicates whether the customer has left the company (Yes/No). This is a binary classification problem.

- Feature Categories:

1. Customer Identification:

****customerID****: Unique identifier for each customer. This feature has no predictive value and will be removed during preprocessing.

2. Demographic Information:

****gender****: Customer gender (Male/Female)

****SeniorCitizen****: Indicates if the customer is a senior citizen (1 = Yes, 0 = No)

****Partner****: Whether the customer has a partner (Yes/No)

****Dependents****: Whether the customer has dependents (Yes/No)

These features may influence customer stability and long-term engagement.

3. Customer Tenure:

****tenure****: Number of months the customer has stayed with the company
This is a key variable, as longer tenure typically indicates higher customer loyalty and lower churn probability.

4. Services Subscribed:

****PhoneService****: Whether the customer has phone service

****MultipleLines****: Whether the customer has multiple lines

****InternetService****: Type of internet service (DSL, Fiber optic, No)

****OnlineSecurity****

****OnlineBackup****

****DeviceProtection****

****TechSupport****

****StreamingTV****

****StreamingMovies****

These features represent the services a customer subscribes to. Customers with more services are generally more engaged and less likely to churn.

5. Contract and Billing Information:

****Contract****: Contract type (Month-to-month, One year, Two year)

****PaperlessBilling****: Whether the customer uses paperless billing

****PaymentMethod****: Payment method used by the customer

Contract type is expected to be one of the strongest predictors of churn, with short-term contracts typically associated with higher churn rates.

6. Financial Features:

****MonthlyCharges****: The amount charged to the customer monthly

****TotalCharges****: The total amount charged to the customer

These features reflect the customer's financial contribution and may influence churn behavior.

- Initial Observations:

1. The dataset contains a mix of categorical and numerical features.
2. Some categorical features contain redundant values such as "No internet service" or "No phone service".
3. The dataset likely contains mild class imbalance in the target variable (Churn).

- Business Understanding:

Customer churn is a critical problem in the telecom industry. Retaining existing customers is often more cost-effective than acquiring new ones. By identifying customers at risk of churning, the company can design targeted retention strategies such as personalized offers, contract adjustments, or improved service plans.

- Expected Key Drivers of Churn:

Based on domain knowledge, the following factors are expected to significantly influence churn:

1. Short contract duration (Month-to-month)
2. Low tenure (new customers)
3. High monthly charges
4. Limited service subscriptions
5. Certain payment methods (e.g., electronic check)